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SUMMARY

AI / GenAI Engineer with 3+ years of experience building production-oriented AI applications using LLMs, VLMs, RAG, multimodal AI, semantic search, and intelligent retrieval. Experienced in developing end-to-end GenAI solutions using LangChain, Azure AI Foundry, Azure AI Search, Azure Machine Learning, Azure Blob Storage, Microsoft Entra ID, FastAPI, LLaMA-3, Mistral, Qwen-VL, LLaVA, SigLIP, BGE-M3, FAISS, and Sentence Transformers. Strong expertise in RAG, LLM-powered query understanding, hybrid and multimodal retrieval, vector search, embeddings, VLM-based image analysis, prompt engineering, and retrieval optimization. Hands-on experience designing secure, scalable AI applications with cloud-based model deployment, API integration, identity and access management, model evaluation, and enterprise AI architectures.

EXPERIENCE

SOFTWARE ENGINEER — Xinthe Technologies (June 2023 – Present)

❖ **Enterprise AI Logo Search & Multimodal Retrieval Platform**

Azure AI Foundry | Azure Machine Learning | Azure AI Search | LangChain | LLMs | VLMs | SigLIP | BGE-M3 | FAISS | SQL Server | Python | FastAPI | Azure Blob Storage | Microsoft Entra ID

- Architected an enterprise-grade multimodal AI search platform for sports/team logo retrieval using LLMs, VLMs, RAG, semantic search, hybrid search, vector search, metadata filtering, and image similarity.
- Designed an end-to-end data ingestion and indexing pipeline covering image preprocessing, VLM-based metadata generation, BGE-M3 text embeddings, SigLIP image embeddings, vector indexing, SQL Server storage, and FAISS similarity indexing.
- Developed VLM-powered image understanding to generate captions, picture types, and structured metadata including symbols, semantic meaning, branding style, emotional attributes, team/fan references, nicknames, and visual characteristics.
- Implemented an LLM-powered query understanding and intent extraction layer using LangChain to identify teams, leagues, entities, locations, colors, sports, semantic attributes, inclusion/exclusion conditions, logical operators, and search requirements.
- Built a hybrid retrieval and RAG pipeline combining lexical/full-text search, metadata filtering, semantic vector search, image similarity, LLM-driven query expansion, candidate generation, multi-stage ranking, and reranking.
- Evaluated and benchmarked LLM/VLM models for query-to-SQL, intent classification, image captioning, metadata extraction, and semantic retrieval, achieving 85–90% accuracy; deployed scalable AI architecture using Azure AI Foundry, Azure ML, Azure AI Search, Blob Storage, Entra ID, FastAPI, and Python.

❖ Enterprise Document Q&A & RAG Platform

FastAPI | LLaMA-3 | Mistral | Ollama | LangChain | RAG | all-MiniLM-L6-v2 | Sentence Transformers | FAISS | Python | SQL

- Developed a production-grade self-hosted GenAI Document Q&A platform using LLaMA-3, Mistral, Ollama, LangChain, FastAPI, and Python, supporting secure offline LLM inference and knowledge-grounded question answering.
- Designed an end-to-end Retrieval-Augmented Generation (RAG) pipeline covering document ingestion, parsing, preprocessing, intelligent text chunking, chunk overlap optimization, embedding generation, vector indexing, semantic retrieval, context construction, and LLM-based response generation.
- Implemented all-MiniLM-L6-v2 / Sentence Transformers embeddings with FAISS vector indexing and similarity search, enabling efficient semantic retrieval of relevant document chunks and context for user queries, with query understanding through keyword extraction, synonym expansion, intent classification, query normalization, and semantic query processing.
- Optimized retrieval and generation using Top-K retrieval, Top-P sampling, similarity thresholds, chunk size, chunk overlap, embedding models, retrieval parameters, prompt engineering, and reranking strategies to improve search relevance, retrieval accuracy, contextual relevance, and answer quality.
- Implemented context-grounded response generation and hallucination reduction, evaluating responses for groundedness, faithfulness, relevance, correctness, contextual precision, answer accuracy, and hallucination behavior based on retrieved document context.
- Built scalable FastAPI REST APIs and asynchronous processing workflows for document ingestion, indexing, query processing, semantic retrieval, context retrieval, LLM inference, and response generation, while evaluating the complete RAG/LLM pipeline for retrieval accuracy, context relevance, answer correctness, grounding, latency, inference performance, and prompting configurations.

❖ Python Backend & AI/ML Application Architecture

Python | FastAPI | Flask | Django | PyTorch | Scikit-learn | Machine Learning | Deep Learning | NLP | Computer Vision | Transformers | Model Training | Model Inference | Model Evaluation | REST APIs | MS-SQL | Async Processing

- Designed and developed scalable Python backend and AI/ML architectures using Python, FastAPI, Flask, and Django, supporting enterprise applications, AI/ML services, model inference, data processing, automation, and production API workloads.
- Developed and integrated Machine Learning and Deep Learning pipelines using PyTorch and Scikit-learn, covering data preprocessing, feature engineering, model training, model evaluation, inference, prediction, classification, and performance optimization for AI-powered applications.
- Built production-ready AI/ML inference and model-serving workflows integrating NLP, Computer Vision, Transformers, LLM/VLM services, preprocessing/post-processing, batch processing, asynchronous execution, and optimized inference pipelines through scalable REST APIs.
- Designed and optimized MS-SQL database schemas, queries, data-access layers, and data pipelines, integrating structured data with AI/ML models, GenAI services, vector search, embeddings, and enterprise backend systems.
- Developed asynchronous and event-driven application workflows with background processing, caching, authentication, middleware, third-party API integration, model deployment, monitoring-ready services, and CI/CD-oriented application architecture, focusing on scalability, reliability, performance, latency optimization, and maintainability.

❖ Computer Vision & Edge AI Architecture

YOLOv5 | OpenCV | OpenVINO | Object Detection | Tracking | Edge AI

- Designed and implemented real-time computer vision architectures for object detection, people counting, surveillance analytics, and video intelligence applications.
- Developed end-to-end video analytics pipelines covering frame extraction, object detection, tracking, ROI-based analysis, event detection, and post-processing.
- Achieved 90%+ counting accuracy in people-counting applications through optimized tracking and post-processing techniques.
- Architected edge AI inference pipelines using Intel OpenVINO, optimizing deep learning models for low-latency and high-FPS deployment scenarios.
- Applied FP16 and INT8 model optimization techniques to improve inference performance and resource utilization on edge devices.
- Integrated computer vision inference services with backend APIs and real-time analytics dashboards, enabling scalable application-level deployment.
- Designed inference workflows with a focus on latency, throughput, hardware utilization, and deployment constraints for real-time computer vision workloads.

❖ Full-Stack & Web Application Engineering

React | JavaScript | PHP | WordPress | HTML | CSS | jQuery

- Designed and developed modular web application architectures integrating frontend interfaces with Python backend services and REST APIs.
- Developed reusable React/JavaScript components and API-driven workflows for interactive enterprise dashboards and business applications.
- Built responsive web interfaces using JavaScript, jQuery, HTML, CSS, and Bootstrap, with emphasis on maintainability and cross-device compatibility.
- Integrated frontend applications with backend APIs, third-party services, and enterprise data workflows to support end-to-end application functionality.
- Developed and customized WordPress-based applications, including themes, reusable UI components, and API integrations.
- Improved application usability and performance through modular frontend architecture, reusable JavaScript components, and optimized data flows.
- Worked across the frontend, backend, API, and database layers, enabling end-to-end ownership of full-stack application workflows.

SOFTWARE ENGINEER INTERN — Xinthe Technologies (Nov 2022 – May 2023)

- Performed QA testing, model validation, inference benchmarking, and performance evaluation for Automatic License Plate Recognition (ALPR) systems using live and recorded video streams, identifying detection and inference issues to improve model reliability.
- Designed and developed real-time people counting and video analytics pipelines using Intel OpenVINO, OpenCV, and computer vision techniques for surveillance and edge AI applications.
- Created and maintained annotated datasets, performed model validation, error analysis, inference output verification, and performance evaluation to improve computer vision model accuracy and deployment readiness.

ACADEMIC PROJECT WORK

Hate Speech Identification using NLP & Machine Learning (Flask)

- Developed an NLP-based hate speech detection system using TF-IDF, Naive Bayes, and Logistic Regression for automated text classification.
- Implemented text preprocessing, feature engineering, sentiment analysis, and model evaluation pipelines to improve prediction accuracy.
- Built a Flask-based web application for real-time content moderation, sentiment visualization, and live text classification.

SKILLS

AI Architecture, GenAI & Agentic AI :

- LLMs, VLMs, Generative AI, Agentic AI, AI Agents, AI Solution Architecture, Multimodal AI
- LangChain, LangGraph, RAG, Prompt Engineering, Tool Calling, Function Calling, Query Understanding, Intent Classification
- LLaMA-3, Mistral, Qwen-VL, LLaVA, Ollama, Hugging Face, Transformers
- Semantic Search, Hybrid Search, Multimodal Retrieval, Image Semantic Search, Information Retrieval, Reranking
- Embeddings, Vector Search, Vector Databases, FAISS, Azure AI Search
- SigLIP, BGE-M3, all-MiniLM-L6-v2, Sentence Transformers
- LLM Evaluation, RAG Evaluation, Grounding, Faithfulness, Relevance, Correctness, Prompt Optimization, Retrieval Optimization
- LLMops, Model Evaluation, Inference Optimization, AI Guardrails, Hallucination Reduction

Machine Learning, Deep Learning & Computer Vision :

- Machine Learning, Deep Learning, PyTorch, Scikit-learn, NLP, Computer Vision, Transformers, HuggingFace
- Model Training, Model Fine-Tuning, Model Inference, Model Evaluation, Feature Engineering, Data Preprocessing
- YOLOv5, OpenCV, OpenVINO, Object Detection, Object Tracking, Video Analytics
- Real-Time Computer Vision, Edge AI, Model Optimization, Inference Optimization, Inference Acceleration
- FP16, INT8, Quantization, GPU/CPU Inference Optimization
- VLM-based Image Understanding, Visual Search, Image Embeddings, Multimodal AI

Azure AI, Cloud & AI Engineering :

- Azure AI Foundry, Azure Machine Learning, Azure AI Search, Azure Blob Storage
- Microsoft Entra ID, Azure App Services, Azure Functions, Managed Endpoints
- Azure OpenAI, AI Model Deployment, Model Lifecycle Management, Cloud AI Architecture
- AI/ML Model Serving, Scalable AI Inference, REST API Integration, Secure AI Application Architecture
- Docker, Linux/Ubuntu, Git, CI/CD, API Deployment

Backend, Data & Production AI :

- Python, FastAPI, Flask, Django, REST APIs, Asynchronous Processing, Background Tasks
- MS-SQL, MySQL, Database Design, Query Optimization, Data Pipelines
- AI/ML Pipelines, Model Serving, Inference Pipelines, API Architecture, Authentication & Authorization
- Production AI Systems, Scalability, Performance Optimization, Latency Optimization, Monitoring & Observability

Backend & API Architecture :

- Python, FastAPI, Flask, Django, REST APIs, API Architecture
- Middleware, Authentication & Authorization, API Integration, Third-Party APIs
- Asynchronous Processing, Background Tasks, Event-Driven Workflows, Task Automation

- AI/ML Model Integration, Model Serving, Inference APIs, Backend-AI Integration
- MS-SQL, MySQL, Database Design, Query Optimization, Data Access Layers

Search, Data & AI Engineering :

- Lexical Search, Full-Text Search, Semantic Search, Hybrid Search, Information Retrieval
- Vector Search, Vector Databases, Metadata Filtering, Similarity Search, Reranking
- Document Processing, Text Chunking, Embedding Pipelines, Vector Indexing, Retrieval Pipelines
- FAISS, Azure AI Search, SQL Server, BGE-M3, all-MiniLM-L6-v2, SigLIP
- Data Pipelines, ETL/ELT, Pipeline Orchestration, Dagster, Data Preprocessing
- Search Relevance, Retrieval Optimization, Query Expansion, Query Processing, Ranking & Reranking

AI Development Tools & IDEs

- OpenAI Codex, Cursor, OpenAI Tools
- AI-Assisted Development, AI Coding Agents, Code Generation, Code Refactoring
- AI-Powered Debugging, Code Analysis, Test Generation, Documentation
- Context-Aware Development, Prompt-Driven Development, Rapid Prototyping
- AI-Assisted Software Engineering, Token-Efficient Development, Developer Productivity

Full-Stack & Application Development :

- ReactJS, JavaScript, React Native, HTML, CSS, Bootstrap, jQuery
- API-Driven Application Development, Frontend-Backend Integration, Enterprise Applications
- Enterprise Dashboards, REST API Integration, Cross-Platform Application Development
- UI Component Development, Responsive Web Applications, Application Performance Optimization

CERTIFICATIONS & TRAINING

- Develop Generative AI Solutions with Azure OpenAI Service – Microsoft Learn
- Develop Generative AI Apps in Azure – Microsoft Learn
- Deploy and Manage Machine Learning Models with Azure Machine Learning – Microsoft Learn
- Explore Azure AI Foundry – Microsoft Learn
- Python Programming – Cisco
- C Language – Cisco
- Dagster Essentials Certification – Dagster University
- Ultimate MS-SQL Bootcamp & Complete JavaScript Course – Udemy

WEBSITES & SOCIAL LINKS

- Portfolio: <https://portfolio-vinodkumargorle.vercel.app/>
- LinkedIn: <https://www.linkedin.com/in/vinodkumargorle8367/>
- GitHub: <https://github.com/vinodkumargorle/>

EDUCATION

- **MCA – Master of Computer Applications (2020–2022)**
Gayatri Vidya Parishad College of Engineering, Visakhapatnam, GPA: **7.6 / 10**
- **B.Sc. – Computer Science (2017–2020)**
Sri GCSR College, Rajam, Andhra Pradesh, GPA: **7.8 / 10**